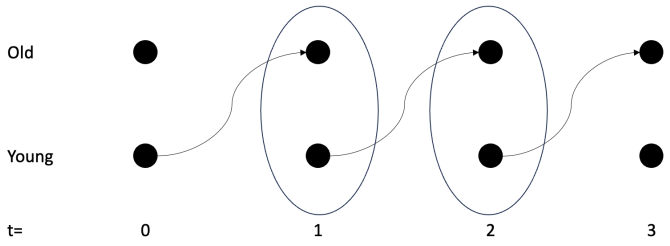
Econ 814 Recitation 5 Notes

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1 Intuition behind OLG model



In the overlapping generations model there is heterogeneity in representative agents. We assume that consumers live for two periods, one where they are young and work, and another where they are old and don't work (these assumptions can be relaxed so that the old can work too). It's overlapping in the sense that there are both old and young in each time period, and when transitioning from one period to the next the young become old and the old die. You could imagine an OLG model with population growth as well, where the number of young is larger than the number of old and so on.

2 OLG comp problem June 23

Consider an overlapping generations economy that goes on to infinity, but in which individuals live only for two periods each. There are many individuals in the economy and we denote the mass of a cohort of workers by L_t. The size of each cohort remains constant over time, and we can normalize $L_t = 1$. There is no production in this economy. Individuals receive exogenous endowments in each period of life. In particular, let y_t denote the endowment of an individual born at time t. The second period endowment of the same individual (born at time t) $y_t(1+\gamma)$, where γ can be negative. Individuals can save through a storage technology, which pays a fixed return $(1+r)$ next period for every unit of the endowment saved this period. An individual born at time t has preferences

over consumption in the two periods of her life:

$$U_t = \log(c_{1,t}) + \beta \log(c_{2,t+1})$$

where $c_{1,t}$ is the consumption of an individual born at time t in the first period of her life, and $c_{2,t+1}$ is the consumption of the same individual in the second period of her life. $\beta \in (0,1)$ is the subjective discount factor. Finally, assume that first period endowments grow at rate between successive generations of young, so that $y_t = (1+\phi)y_{t-1}$.

2.1 Part a

Write down the resource constraint for this economy at time t.

The resource constraint for this economy would be the aggregate consumption plus investment is equal to the aggregate endowment

$$c_t + s_t = y_t$$

2.2 Part b

Write down the budget constraints for an individual born at time t. Assuming that the individual maximizes utility, solve for her choice of first period consumption and savings rate, $s_{rt} = s_t/y_t$.

The budget constraints for individuals born in period t are:

$$c_{1,t} + s_t = y_t$$

$$c_{2,t+1} = y_t(1+\gamma) + s_t(1+r)$$

To solve for the optimal first period consumption we can set up the maximization problem and substitute the budget constraints to avoid a constrained maximization problem.

$$\max_{c_{1,t}, c_{2,t+1}} \log(c_{1,t}) + \beta \log(c_{2,t+1})$$

$$\max_{s_t} \log(y_t - s_t) + \beta \log(y_t(1+\gamma) + s_t(1+r))$$

FOC for s_t

$$\frac{-1}{y_t - s_t} + \frac{\beta(1+r)}{y_t(1+\gamma) + s_t(1+r)} = 0$$

Solve to find that

$$s_t = y_t \frac{\beta(1+r) - (1+\gamma)}{(1+\beta)(1+r)}$$

$$s_{rt} = s_t/y_t = \frac{\beta(1+r) - (1+\gamma)}{(1+\beta)(1+r)}$$

Plug in to find the first period consumption

$$c_{1,t} = y_t - s_t = y_t(1 - \frac{\beta(1+r) - (1+\gamma)}{(1+\beta)(1+r)})$$

2.3 Part c

How does the growth of income over the life-cycle, γ, expected by one individual affect her savings rate? Explain. How does an increase in ϕ affect the growth of aggregate savings in this economy? Explain.

If γ increases, this means the savings rate decreases. This makes sense, since if the amount of endowment increases next period then the consumer can respond by consuming more today. If ϕ increases, the growth rate of the aggregate savings increases since y_t would increase.

3 OLG comp problem June 22

Overlapping Generations model with human capital.

Consider an overlapping generations model where agents live for three periods. At each date N households are born. When young in period $t-1$, a household borrows an amount d_{t-1} at the market rate of return to acquire human capital, h_t, according to $h_t = d_{t-1}^{\rho}$ with $0 < \rho < 1$.

When middle-aged, in period t, the household repays the debt accumulated, earns labor income $w_t h_t$ where w_t is the wage rate, repays the debt accumulated in the previous period at the gross real interest rate r_t, consumes $c_{1,t}$, and saves s_t. When old, in period $t+1$, the household does not work and consumes $c_{2,t+1}$. Preferences for an agent born in period $t-1$ are given by:

$$U(c_{1,t}, c_{2,t+1}) = \log c_{1,t} + \log c_{2,t+1}$$

Firms maximize profits taking prices as given using the production function

$$F(K_t, L_t) = K_t^{\alpha} L_t^{1-\alpha}$$

Capital, which accumulates according to $K_{t+1} = I_t$ (so there is full depreciation) and $L_t = N h_t$

3.1 Part a

Write the optimization problem for an agent born in period $t-1$ and derive the corresponding first-order conditions.

Let's first define the budget constraints. In the middle age period, the consumer can either consume, save, or pay off its debt out of its income.

$$c_{1,t} + s_t + r_{t-1} d_{t-1} = w_t h_t$$

When old, the consumer consumes their savings[1]

$$c_{2,t+1} = r_t s_t$$

Now we can set up the maximization

$$\max_{c_{1,t}, c_{2,t+1}, s_t, d_{t-1}} \log c_{1,t} + \beta \log c_{2,t+1}$$

Substitute the constraints

$$\max_{s_t, d_{t-1}} \log(w_t d_{t-1}^{\rho} - s_t - r_{t-1} d_{t-1}) + \beta \log(r_t s_t)$$

FOC s_t

$$\frac{-1}{w_t d_{t-1}^{\rho} - s_t - r_{t-1} d_{t-1}} + \frac{\beta r_t}{r_t s_t} = 0$$

Solving to find

$$s_t = \frac{\beta w_t d_{t-1}^{\rho} - r_{t-1} d_{t-1}}{1+\beta}$$

FOC d_{t-1}

$$\frac{w_t \rho d_{t-1}^{\rho-1} - r_t}{w_t d_{t-1}^{\rho} - r_{t-1} d_{t-1} - s_t} = 0$$

Solve to find

$$d_{t-1} = \left(\frac{r_t}{\rho w_t}\right)^{\frac{1}{\rho-1}}$$

3.2 Part b

Write an agents savings, s_t, and debt, d_{t-1}, as a function of variables exogenous to the agent.

We already have debt, now for savings substitute the expression for debt

$$s_t = \frac{\beta w_t \left(\frac{r_t}{\rho w_t}\right)^{\frac{\rho}{\rho-1}} - r_{t-1}\left(\frac{r_t}{\rho w_t}\right)^{\frac{1}{\rho-1}}}{1+\beta}$$

3.3 Part c

Define a competitive equilibrium for this economy

A CE for this OLG economy is a sequence of prices $\{r_t, w_t\}$ allocations for the consumer $\{c_{1,t}, c_{2,t+1}, s_t, h_t, d_{t-1}\}$ and allocations for the firm $\{K_t, L_t\}$ such that given prices, the allocation for the consumer solves the young cosnumers problem, the allocation for the firm solves the firms problem, and markets clear such that $L_t = N h_t$, $S_t = K_{t+1}$ and $c_{1,t} + c_{2,t} + s_t = F(K_t, L_t)$.

[1] Note that when I presented this in recitation I had a $(1 + r_t)$ here. This was incorrect since there is full depreciation in the economy the old consumers should only consume out of the appreciated savings.

3.4 Part d

Derive the equations that describe the dynamics of aggregate physical and human capital per worker.

Let's start with savings:

We have that $S_t = K_{t+1}$ but we want the per worker savings from above

$$s_t = K_{t+1}/N = \frac{K_{t+1}}{L_{t+1}} \frac{L_{t+1}}{N} = k_{t+1} h_{t+1}$$

$$s_t = \frac{\beta w_t (\frac{r_t}{\rho w_t})^{\frac{\rho}{\rho-1}} - r_{t-1} (\frac{r_t}{\rho w_t})^{\frac{1}{\rho-1}}}{1+\beta}$$

Plug in w_t, r_t from the firm's problem, with $K/N = k$

$$k_{t+1} h_{t+1} = \frac{\beta(1-\alpha) k_t^{\alpha} (\frac{\alpha}{(1-\alpha)k_t})^{\frac{\rho}{\rho-1}} - \alpha k_{t-1}^{\alpha-1} (\frac{\alpha}{(1-\alpha)k_t})^{\frac{1}{\rho-1}}}{1+\beta}$$

So we have that capital per worker is a function of constants, state variable k_t, and human capital h_{t+1}.

$$h_{t+1} = d_t^{\rho}$$

And from above we have

$$d_{t-1} = (\frac{r_t}{\rho w_t})^{\frac{1}{\rho-1}}$$

So plugging in from the firm's problem

$$h_{t+1} = (\frac{\alpha}{\rho(1-\alpha)k_{t+1}})^{\frac{\rho}{\rho-1}}$$

Plug in for the h_{t+1} in equation above

$$k_{t+1} (\frac{\alpha}{\rho(1-\alpha)k_{t+1}})^{\frac{\rho}{\rho-1}} = \frac{\beta(1-\alpha) k_t^{\alpha} (\frac{\alpha}{(1-\alpha)k_t})^{\frac{\rho}{\rho-1}} - \alpha k_{t-1}^{\alpha-1} (\frac{\alpha}{(1-\alpha)k_t})^{\frac{1}{\rho-1}}}{1+\beta}$$

And simplify to solve for k_{t+1}

$$k_{t+1} = [(\frac{\alpha}{\rho(1-\alpha)})^{\frac{\rho-1}{\rho}} \frac{\beta(1-\alpha) k_t^{\alpha} (\frac{\alpha}{(1-\alpha)k_t})^{\frac{\rho}{\rho-1}} - \alpha k_{t-1}^{\alpha-1} (\frac{\alpha}{(1-\alpha)k_t})^{\frac{1}{\rho-1}}}{1+\beta}]^{1-\rho}$$